4.5 Oscillations

This topic covers simple harmonic motion and damping.

This topic may be taught using applications that relate to, for example, the construction of buildings in earthquake zones.

Students will be assessed on their ability to:		Suggested experiments
119	recall that the condition for simple harmonic motion is $F = -kx$, and hence identify situations in which simple harmonic motion will occur	
120	recognise and use the expressions $a = -\omega^2 x$, $a = -A\omega^2 \cos \omega t$, $v = -A\omega \sin \omega t$, $x = A\cos \omega t$ and $T = 1/f = 2\pi/\omega$ as applied to a simple harmonic oscillator	
121	obtain a displacement–time graph for an oscillating object and recognise that the gradient at a point gives the velocity at that point	Use a motion sensor to generate graphs of SHM
122	recall that the total energy of an undamped simple harmonic system remains constant and recognise and use expressions for total energy of an oscillator	
123	distinguish between free, damped and forced oscillations	
124	investigate and recall how the amplitude of a forced oscillation changes at and around the natural frequency of a system and describe, qualitatively, how damping affects resonance	Use, for example, vibration generator to investigate forced oscillations
125	explain how damping and the plastic deformation of ductile materials reduce the amplitude of oscillation	Use, for example, vibration generator to investigate damped oscillations